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UNIVERSITI TEKNOLOGI MALAYSIA

UNIVERSITI TEKNOLOGI MALAYSIA
FACULTY OF MECHANICAL ENGINEERING

PROJECT REPORT

MEMM1263: CONDITION MONITORING

GROUP MEMBER :

1	HU DEZHI	MEM 241002
2	NEOH ZONG YU	MEM 241008
3	NIU TIEZHU	MEM 231069
4	SUM JING YU	MEM 241009

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PROJECT TOPIC : WIRELESS VIBRATION MONITORING SYSTEM FOR
TABLE FAN OPERATION

LECTURER : PROF. MADYA. DR. ZAIR ASRAR BIN AHMAD

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Rotating machinery such as motors, pumps, and fans are critical components in industrial operations, where unexpected failures can result in costly downtime, reduced efficiency, and safety hazards. Traditionally, condition monitoring relies on manual inspections or wired sensors, which are often expensive, labor-intensive, and lack real-time accessibility. As a result, there is an increasing need for cost-effective, wireless solutions that enable continuous and remote monitoring of equipment health. Wireless vibration monitoring presents a promising alternative by leveraging low-power microcontrollers and MEMS sensors to detect early fault signatures such as unbalance, misalignment, and bearing defects. Advances in embedded processing and cloud computing now allow real-time signal processing techniques like Fast Fourier Transform (FFT) and statistical indicators to be deployed on edge devices. These methods enhance fault detection precision and provide predictive maintenance insights.

The integration of wireless communication technologies with cloud-based dashboards empowers operators to visualize vibration data trends, receive automated alerts, and take timely corrective actions. This project aims to develop a compact, affordable, and efficient wireless vibration monitoring system that incorporates fault detection algorithms and a user-friendly dashboard, bridging the gap between industrial reliability needs and accessible smart monitoring solutions.

1.2 Problem Statement

Despite the increasing need for predictive maintenance, small- and medium-sized industries often lack access to affordable and scalable vibration monitoring systems. Existing commercial solutions are either too expensive, require complex installation, or lack wireless capabilities, making them unsuitable for widespread deployment, especially in retrofitting applications. Moreover, many traditional systems do not provide real-time data analysis or automated alerts, limiting their ability to support proactive maintenance decisions. As a result, machinery faults such as unbalance, misalignment, or bearing degradation may go undetected until significant damage occurs, leading to downtime, reduced efficiency, and higher operational costs.

Therefore, there is a critical need to develop a low-cost, wireless, and real-time vibration monitoring system capable of detecting early-stage mechanical faults using signal processing techniques like FFT and statistical analysis. This system should also integrate with cloud-based dashboards to enable remote data visualization and alerting, thereby supporting smarter, data-driven maintenance strategies.

1.3 Project Objective

The objective of this project is shown below:

1. To develop a wireless vibration monitoring system for table fan operation

1.4 Project Scope

This project aims to develop a wireless vibration monitoring system using an ESP32 microcontroller, an MPU6050 accelerometer, and a portable power source. The

system will capture real-time vibration signals, apply signal processing techniques such as FFT, RMS and kurtosis approach for early fault detection, and transmit data wirelessly via Wi-Fi or LoRa. A cloud-based dashboard will visualize vibration trends and frequency spectra, with automated alerts sent when abnormal conditions are detected. The project will be fully documented through a presentation, demonstration video, and an open-source GitHub repository containing all code, designs, and datasets.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter reviewed the fundamental concepts of vibration, including key terminology and behavior. It also highlighted the negative impacts of vibration on rotating machinery, with a particular focus on table fans, which are commonly found in households. Common faults associated with table fans and electric fan motors, as well as existing fault detection methods, were briefly explored. Finally, the role of the Internet of Things (IoT) in the implementation of vibration monitoring systems was discussed.

2.2 Fundamental Concepts and Terminology of Vibration

Vibration can be defined as a repetitive, periodic, or oscillatory motion exhibited by mechanical systems, particularly in rotating machinery [1]. The number of vibration cycles per second is commonly referred to as "frequency." When these repetitive motions are smooth, regular, and occur at relatively low frequencies, they are often referred to as "oscillations." In contrast, motions that happen at higher frequencies, with lower amplitudes and irregular or random behavior, are generally classified as vibrations. Despite these distinctions, the terms "vibration" and "oscillation" are often used interchangeably. Vibrations can occur naturally in engineering systems, reflecting their inherent dynamic behavior, or they can be the result of external or internal excitations. These excitation forces may arise from within the system or be applied from outside. When the frequency of an external force matches the system's natural frequency, the

system experiences a significant increase in vibration amplitude, a phenomenon known as "resonance" [2]. The corresponding frequency is called the "resonant frequency." Resonance occurs when a system's operating frequency aligns with its natural frequency. Vibrations can be either beneficial or harmful. In some cases, controlled vibrations serve useful purposes, while in others, excessive or unwanted vibrations can lead to discomfort, damage, or system failure. In many engineering applications, operating at resonance is undesirable due to the potential for destructive effects. Therefore, understanding human responses to vibration, suppressing harmful vibrations, and promoting beneficial ones are key objectives in the field of vibration engineering.

Mechanical systems with reciprocating or rotating components inherently generate vibrations due to disturbances from sources such as combustion engines, structural impacts, or aerodynamic forces. In mechanical engineering, free vibration specifically refers to oscillatory motion that occurs without external forcing. On the other hand, uncontrolled vibrations can compromise machinery performance, operational efficiency, and safety, ranging from minor disruptions to catastrophic failures. As a critical parameter in general health monitoring, machine vibration analysis gained significant attention for its role in improving reliability, lifespan, quality control, and failure prevention. Overall, the vibrations are quantified using three key parameters, as displayed in Table 2.1, while the sensor selection for measurement depends on frequency response, as sensitivity varies across the vibration spectrum.

Table 2.1 : Parameters used to quantify the vibration [3]

No.	Parameter	Description
1	Displacement / Amplitude	Peak position at the end of the motion
2	Velocity	Rate of motion
3	Acceleration	Rate of velocity change

2.3 Vibration Measuring Method

The vibrations produce measurable physiological and psychological effects on humans. From a physiological perspective, these effects necessitate rigorous vibration testing and monitoring protocols. Industry standards and government regulations strictly define permissible vibration levels, frequency ranges, and exposure durations across different applications. In order to develop an effective vibration monitoring system, several key components are required for implementation, as listed in Table 2.2.

Table 2.2: Key components for measuring the vibration

No.	Components	Description
1	Sensor and Transducer	• Tool used to convert mechanical vibration into an electrical signal
2	Signal Conditioning Hardware	• Used for filtering the data noise/denoising • Used for amplifying or boosting the signal • Modulator for signal processing • Analog to digital converter for data acquisition

To measure machinery vibration, a transducer or sensor is used, which converts changes in mechanical quantities into changes in other physical quantities, usually an electrical signal proportional to a parameter of the experienced motion. Today, there are three commonly used transducers for vibration measurement which are displacement sensors, velocity sensors, and accelerometers [4]. Each type of sensor has its advantages and limitations, which vary depending on the application. The selection of a suitable sensor is primarily based on factors such as frequency range, sensitivity, and operating conditions. While emerging approaches such as using vision data from event-based cameras are gaining attention, accelerometers remain the most commonly used sensors due to their higher accuracy, wide measurement range, ease of installation, and cost-effectiveness. Additionally, acceleration signals can be easily integrated numerically to

obtain velocity and displacement. Figure 2.1 shows the basic principle of vibration measurement.

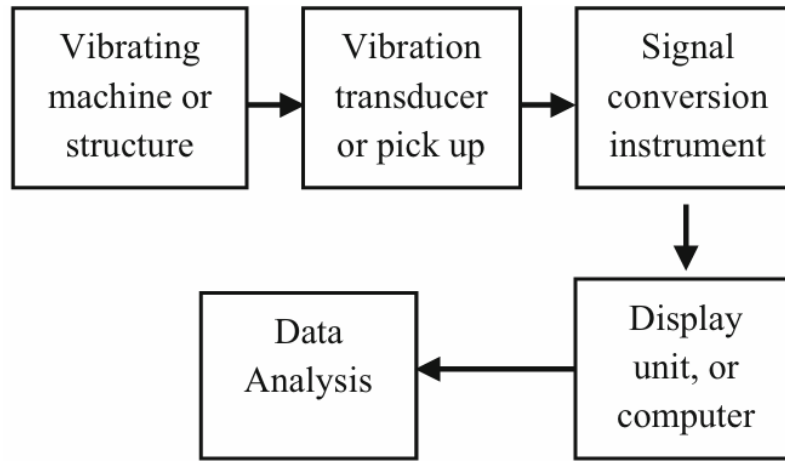


Figure 2.1 : Principle of vibration measurement [5]

2.4 Vibration Analysis Method

The analysis of a vibrating system can be performed either in the time domain or the frequency domain. It is important to note that the data obtained from sensors or transducers is inherently time-domain-based, as these devices measure physical motion in real time, capturing how vibration intensity varies from moment to moment. In the time domain, the independent variable of a vibration signal is time, and the system itself can be modelled as a set of time-dependent differential equations. However, the use of time-domain data in vibration analysis is often limited, as it fails to reveal critical information such as the specific frequencies causing the vibration, hidden harmonic patterns, and resonance frequencies. Therefore, a transfer function is used to convert time-domain data into the frequency domain

The time and frequency domains are interconnected through the Fourier transform, which can be viewed as a special case of the Laplace transform. These transformation techniques are first examined in the context of purely analytical and analogue continuous-

time systems. In practical applications, however, digital electronics and computers are commonly employed for signal analysis, sensing, and control. In this context, the Fast Fourier Transform (FFT), which is an efficient algorithm for computing the Discrete Fourier Transform (DFT), is widely used to convert time-domain vibration signals into frequency-domain spectra. The DFT equation can be derived as [6]:

$$X[k] = \sum_{n=0}^{N-1} x[n] \cdot e^{\frac{-2j\pi kn}{N}} \quad (2.1)$$

Where $X[k]$ = Frequency-domain output
 $x[n]$ = Time-domain sample

The DFT requires computing the sum of products for each of the n frequency components across n time-domain samples. If calculating the discrete Fourier transform of a signal, it would need to perform N (multiplications) $\times$ N (additions) = $O(N^2)$ operations. Then, the FFT revolutionizes this process through algorithmic optimization. As its name suggests, the FFT dramatically accelerates DFT computation by exploiting mathematical symmetries and divide-and-conquer strategies. This optimization reduces the computational complexity from $O(N^2)$ to just $O(N \log N)$, which is a game-changing improvement that makes spectral analysis practical for real-time applications.

In vibration analysis, the time domain represents vibration signals as a function of time. Time-domain analysis commonly uses the Root Mean Square (RMS) value to quantify the overall energy of the vibration signal, capturing amplitude variations over time. The primary advantage of time-domain analysis lies in its simplicity, as a single RMS value can indicate the overall severity of vibration, where an increasing RMS level typically signals the deterioration of machine condition. This makes it valuable for tracking changes in machine health and detecting sudden impacts or shocks during operation. In contrast, frequency-domain analysis is essential for uncovering hidden patterns that are critical for machinery diagnostics. By isolating vibration components

based on frequency, this approach helps identify the specific source of defects and detect characteristic fault frequencies. Additionally, frequency-domain analysis can reveal natural frequencies that must be avoided to prevent resonance and potentially destructive operation.

2.5. Time-Domain RMS Value Specification

The root mean square value of a vibration signal is given by the square root of the average of the square signal, which is derived as below [7]:

$$RMS = \sqrt{\frac{1}{N} * \sum_{i=1}^N x_i^2} \quad (2.2)$$

Where x_i = Instantaneous amplitude of the vibration signal at the sample i
 N = Total number of samples in the window

The RMS value commonly serves as a fundamental metric in time-domain vibration analysis, providing a statistically significant measure of a signal's effective energy content. Mathematically defined as the square root of the arithmetic mean of squared instantaneous amplitudes over a specified time window, RMS quantifies the continuous power-equivalent magnitude of oscillatory vibration signals. This parameter offers particular utility in mechanical systems monitoring as it inherently accounts for both the amplitude and duration of vibrational energy, unlike peak measurements, which only capture extreme instantaneous values. The RMS value's direct correlation with vibration energy makes it indispensable for assessing overall machine health, detecting progressive degradation patterns, and evaluating compliance with international standards such as ISO 10816 for vibration severity classification. In practical applications, trending RMS values over time enables early identification of developing faults, where increasing

RMS levels typically indicate worsening mechanical conditions, whether from imbalance, misalignment, bearing wear, or other common failure modes. The metric's robustness against transient spikes and electrical noise further enhances its reliability for condition monitoring in industrial environments.

2.6 Role of Internet of Things (IoT) in Condition Monitoring

Today, vibration condition monitoring is vital for detecting early signs of mechanical faults in rotating machinery or equipment, such as motors, pumps, compressors, and fans. However, traditional vibration monitoring systems are often offline, expensive, and require manual data collection. Therefore, this is where the Internet of Things (IoT) revolutionizes the approach. The IoT represents a network of interconnected physical devices, including sensors, machinery, and appliances, that communicate through internet protocols to enable real-time data collection, transmission, and analysis. In conventional condition monitoring systems, the essential architecture typically comprises three key elements: sensors that capture operational parameters, a microcontroller unit responsible for data acquisition and processing, and a communication module that facilitates wireless data transmission to cloud-based systems or local devices via Wi-Fi, Bluetooth, or other IoT connectivity protocols.

The implementation of IoT enables continuous, automated vibration data collection without manual intervention, while facilitating remote monitoring through wireless data transmission. This capability enhances data accessibility across teams and systems. By leveraging real-time vibration analysis, the system improves fault detection accuracy and triggers instant alerts through any online platform when the vibrations from the rotating equipment exceed the predefined thresholds. Consequently, this approach allows the transition of maintenance strategy from reactive to predictive maintenance strategies, significantly reducing unplanned downtime and optimizing operational efficiency.

CHAPTER 3

RESEARCH METHODOLOGY

3.1 Introduction

This chapter presents the experimental setup for the development of a wireless vibration condition monitoring system for a table fan. The methodology is divided into three main categories: hardware setup, software setup, and boundary condition setup, which refers to the simulation of the testing conditions.

3.2 Hardware Development and Setup

3.2.1 Selection of Hardware Components

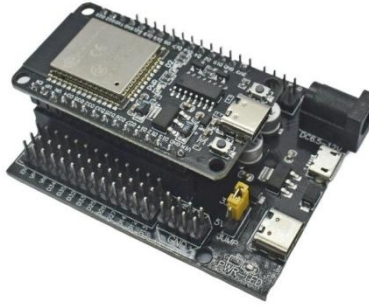
To develop a wireless vibration condition monitoring system, the use of a sensor or transducer is essential for converting mechanical vibrations into electrical signals. A microcontroller is then required to acquire this sensor data and perform post-processing using predefined algorithms. Additionally, the microcontroller must be equipped with a Wi-Fi communication module to enable wireless data transmission based on programmed instructions. All these components must be seamlessly integrated, much like the interconnectedness of blood vessels in the human body. Therefore, the hardware components used to construct the wireless vibration monitoring system for the table fan are listed in Table 3.1, while Figure 3.1 illustrates these components.

Table 3.1: Hardware component list with description

No.	Component	Quantity
1	MPU 6050 Accelerometer	1
2	NodeMCU-ESP32	1
3	Jumper Wire	4



(a)



(b)



(c)

Figure 3.1 : (a) MPU6050 6DOF Accelerometer [8]; (b) NodeMCU-ESP32 [9]; (c) Jumper wire [10]

Basically, the MPU6050 accelerometer is an integrated sensor module that combines a 3-axis accelerometer and a 3-axis gyroscope, enabling it to measure both linear acceleration and rotational motion. The NodeMCU-ESP32, on the other hand, serves as the central microcontroller and data processing unit of the system. It also manages wireless communication through its built-in Wi-Fi module, allowing processed data to be transmitted to the cloud platform for remote monitoring and visualization. Finally, jumper cables provide physical connection between the MPU6050 sensor and the NodeMCU-ESP32 to complete the hardware assembly. Once connected, the microcontroller will operate wirelessly to transmit vibration data without relying on physical data transfer.

3.2.2 Sensor Placement Positioning

To collect vibration data from the table fan, the MPU6050 sensor was positioned near the electric motor side, where vibrations are most prominent. Proper sensor placement is crucial to ensure the alignment of the sensor's X, Y, and Z axes with the principal directions of vibration, typically corresponding to the horizontal, vertical, and axial directions of the machine. Figure 3.2 illustrates the vibration detection directions of the MPU6050 sensor. Accordingly, the sensor was arranged to capture vibrations from the correct orientation. In this project, the MPU6050 was positioned such that the X-axis measures axial vibrations along the fan rod or shaft axis, the Y-axis measures horizontal vibrations perpendicular to the X-axis, and the Z-axis captures vertical vibrations. Figure 3.3 summarizes the MPU6050 axis arrangement on the table fan and the actual placement of the sensor on the fan casing. Finally, the accelerometer sensor was securely fixed onto the table fan's casing using hot glue. This method ensures that the sensor remains firmly attached with minimal damping effects that could lead to data collection errors. Additionally, the use of hot glue helps maintain proper alignment of the sensor's axes, which is particularly important when mounting the sensor on the curved surface of the table fan casing.

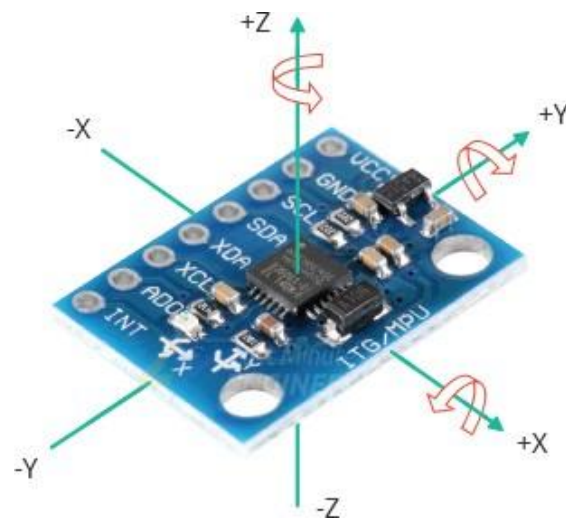


Figure 3.2 : MPU 6050 sensing direction for alignment purposes [11]

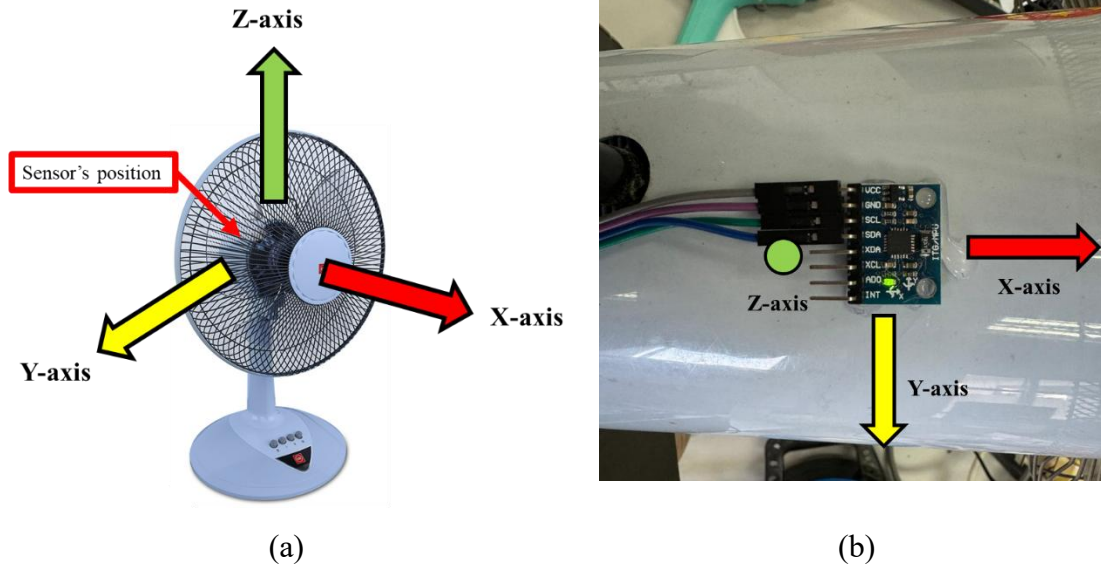


Figure 3.3: (a) Demonstration of pre-select table fan axis: (b) Actual axis setup for sensor on the table fan's casing

3.2.3 Power Supply and Circuit Integration

This section provides an overview of how the hardware components are interconnected. The MPU6050 sensor is connected to the NodeMCU ESP32 using wired connections, as illustrated in Figure 3.4 with the usage of Fritzing simulation software. On the other hand, a simplified circuit diagram showing the hardware connections and communication pathways is presented in Figure 3.5, where the NodeMCU supplies power to the MPU6050 through the VCC and GND lines and communicates via the SCL and SDA pins using the I²C protocol. The overall system consists of four main components: the laptop, NodeMCU-ESP32, MPU6050 sensor, and ThingSpeak cloud platform. The laptop supplies power to the NodeMCU via a USB cable, while the NodeMCU transmits data wirelessly to ThingSpeak over Wi-Fi. Notifications such as email alerts are received on the laptop, which will be further discussed in the following section.

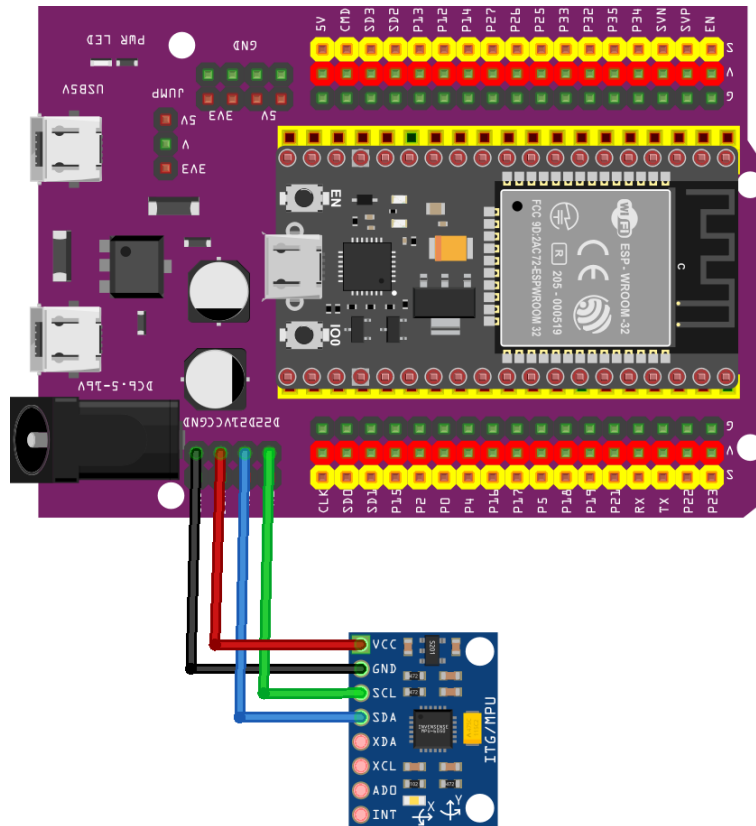


Figure 3.4: Wired connection diagram between MPU 6050 and NodeMCU ESP 32

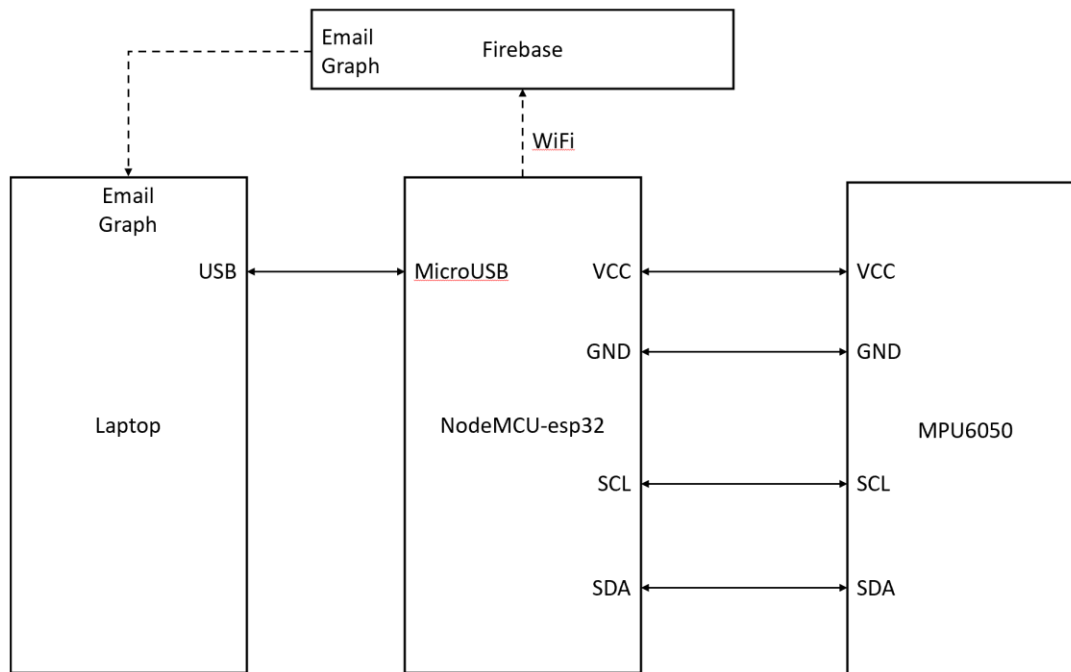


Figure 3.5: Overall circuit diagram of the system

3.3 Data Analysis and Fault Detection

3.3.1 Data Selection and Preprocessing

This stage is crucial for ensuring the acquisition of high-quality vibration data prior to analysis. To achieve this, the output range of the MPU6050 accelerometer is configured to $\pm 8g$ using the command `mpu.setAccelerometerRange(MPU6050_RANGE_8_G)`, as shown in Figure 3.6 within the Arduino code. Setting the sensor to this range allows it to capture a wide spectrum of vibration amplitudes typically encountered in industrial applications without signal saturation or clipping. This configuration ensures that both low-level vibrations and higher-impact events can be accurately recorded, providing reliable input for subsequent signal processing and fault detection. The data preprocessing configuration uses a 1,000 Hz sampling rate with a 256-sample buffer to optimize vibration analysis while balancing key technical constraints, as displayed in Figure 3.7.

```
// Set accelerometer range to  $\pm 8g$  (adjustable)
mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
Serial.print("Accelerometer range set to  $\pm$ ");
Serial.println(mpu.getAccelerometerRange() == MPU6050_RANGE_8_G ? "8G" : "Unknown");
```

Figure 3.6: Setting for configuring MPU6050's output range

```
// ==== Configuration Constants ====
#define SAMPLE_RATE 1000           // Hz
const uint16_t NUM_SAMPLES = 256; // Must be power of 2 for FFT
```

Figure 3.7: Setting for sampling rate and buffer size

This setup adheres to the Nyquist theorem, which requires sampling at least twice the highest frequency component of interest, while in this case, enabling reliable capture of vibrations up to 500 Hz. The 256-sample buffer represents a careful compromise between frequency resolution and system responsiveness. While smaller buffers would

provide faster updates, they would significantly degrade frequency resolution, potentially masking important spectral features. Conversely, larger buffers would improve resolution but introduce problematic latency in real-time monitoring applications. The chosen 256-sample configuration delivers 3.9 Hz frequency resolution while maintaining reasonable computational efficiency on the ESP32 platform, making it particularly suitable for detecting common machinery faults like imbalance and misalignment that typically manifest in the sub-500 Hz range. This balanced approach ensures effective vibration monitoring without overburdening the microcontroller's processing capabilities.

3.3.2 Time-Domain Analysis with RMS and Kurtosis

Time-domain analysis processes raw vibration signals, which represent vibration amplitude over time, to extract key statistical metrics for fault detection. One such metric is the Root Mean Square (RMS) value, which quantifies the overall vibration energy and correlates directly with the severity of mechanical faults. For additional detail, the RMS value is separated into three components corresponding to the X, Y, and Z axes of the sensor. By comparing the calculated RMS value against predefined thresholds, the system can detect general mechanical issues such as looseness or imbalance. In addition to RMS, kurtosis is implemented to identify non-Gaussian, impulsive vibration patterns typically associated with localized defects like bearing spalls or gear tooth impacts. Figure 3.8 illustrates the configuration for time-domain RMS calculation, while Figure 3.9 shows the kurtosis implementation within the Arduino code.

```
// Time-domain RMS
for (int i = 0; i < NUM_SAMPLES; i++)
{
    sumSq += real[i] * real[i];
    mean += real[i];
}
float rms_full = sqrt(sumSq / NUM_SAMPLES);
mean /= NUM_SAMPLES;
```

Figure 3.8: Time-domain RMS in Arduino code

```
float kurtosis = (variance > 0) ? (fourthMoment / (variance * variance)) : 0;
```

Figure 3.9: Kurtosis in Arduino code

3.3.3 Frequency-Domain Analysis (FFT)

The time-domain vibration signals undergo a transformation using the FFT method, which converts them into the frequency domain, revealing the periodic components and their corresponding amplitudes. Figure 3.10 illustrates the FFT algorithm implemented in the code that performs this transformation from time-domain to frequency-domain data.

```
// FFT objects for each axis
ArduinoFFT<double> FFT_X(vRealX, vImagX, NUM_SAMPLES, SAMPLE_RATE);
ArduinoFFT<double> FFT_Y(vRealY, vImagY, NUM_SAMPLES, SAMPLE_RATE);
ArduinoFFT<double> FFT_Z(vRealZ, vImagZ, NUM_SAMPLES, SAMPLE_RATE);
```

Figure 3.10: Fast Fourier Transform for each axis of data

Thus, vibration analysis can be performed in the frequency domain. The frequency-domain RMS is used to quantify the energy content of the signal at specific fault frequencies, which are often associated with mechanical issues such as bearing defects, gear mesh problems, or misalignment. Unlike the time-domain RMS, which reflects the overall vibration energy, the frequency-domain RMS isolates and highlights the energy concentrated at particular frequencies of interest, making it more effective for diagnosing specific faults. In addition, peak frequency detection is implemented as a complementary technique that identifies the frequency bin with the highest amplitude within a specified range. This method is particularly useful for detecting dominant vibration components that may indicate faults, resonances, or abnormal conditions. Figure 3.11 presents the code implementation for frequency-domain RMS, while Figure 3.12 shows the code used for peak frequency detection.

```
// Frequency-domain analysis
fft.windowing(FFT_WIN_TYP_HAMMING, FFT_FORWARD);
fft.compute(FFT_FORWARD);
fft.complexToMagnitude();
```

Figure 3.11: Frequency-domain RMS in Arduino code

```
// peak
double peak = 0;
int peakIndex = -1;
int maxBin = (int)(100.0 * NUM_SAMPLES / SAMPLE_RATE); // limits the peak search to ~100 Hz

for (int i = 1; i < maxBin; i++)
{
    if (real[i] > 0.01 && real[i] > peak) // skips FFT bins with very low magnitude (likely noise)
    {
        peak = real[i];
        peakIndex = i;
    }
}
```

Figure 3.12: Peak frequency detection in Arduino code

3.3.4 Threshold Setting and Fault Diagnosis

Threshold values play a key role in automating fault detection in machinery by flagging abnormal vibration patterns. The RMS value, which measures the overall energy level of a vibration signal in the time domain, is commonly used for this purpose. A high RMS value indicates increased vibration severity, often associated with mechanical faults such as unbalance, misalignment, or bearing wear. In this project, it was observed that even during normal operation of the table fan at a constant RPM, the RMS value consistently exceeded 0.5g. This is likely due to the fan's lightweight components and less rigid structure, which make it more sensitive to minor imbalances. Therefore, the RMS threshold was increased to a higher value to avoid unnecessary false alarms, ensuring that only significant issues such as severe imbalance, broken blades, or loose mounts trigger fault detection. Similar to the RMS threshold, the kurtosis threshold is set to 6.0, which is

used to identify when a vibration signal deviates significantly from a normal (Gaussian) distribution, potentially indicating the presence of faults or abnormal operating conditions.

```
// Threshold values for fault detection
const float RMS_THRESHOLD = 3;          // RMS warning
const float KURTOSIS_THRESHOLD = 6.0;  // Kurtosis > 6.0 indicates impulsive vibration
```

Figure 3.13: Threshold value setup in Arduino code

3.4 Cloud Dashboard and Alert System

3.4.1 Cloud Platform Selection

In this project, ThingSpeak has been selected as the cloud platform to enable the Internet of Things (IoT) functionality. The acceleration data generated by the MPU6050 sensor is transmitted wirelessly by the NodeMCU ESP32 to ThingSpeak via Wi-Fi and visualized in a customized channel named “1263 Project”, as shown in Figure 3.14. A total of seven fields are used to display key vibration parameters: RMS @ 1×RPM (X-axis), Frequency @ 1×RPM (X-axis), RMS @ 1×RPM (Y-axis), Frequency @ 1×RPM (Y-axis), RMS @ 1×RPM (Z-axis), Frequency @ 1×RPM (Z-axis), and the device’s geographical location on a map, as display in Figure 3.15. Through this cloud visualization, operators can easily monitor vibration trends over time and take timely action to support predictive maintenance strategies. Additionally, the vibration data is not only visualized on ThingSpeak but also used to trigger notifications via Wi-Fi, as described in the previous section.

My Channels

[New Channel](#)

Search by tag

Name

1263 project

Private Public Settings Sharing API Keys Data Import / Export

Created2025-06-10

Updated2025-06-16 03:45

Figure 3.14 : Customized channel in ThingSpeak

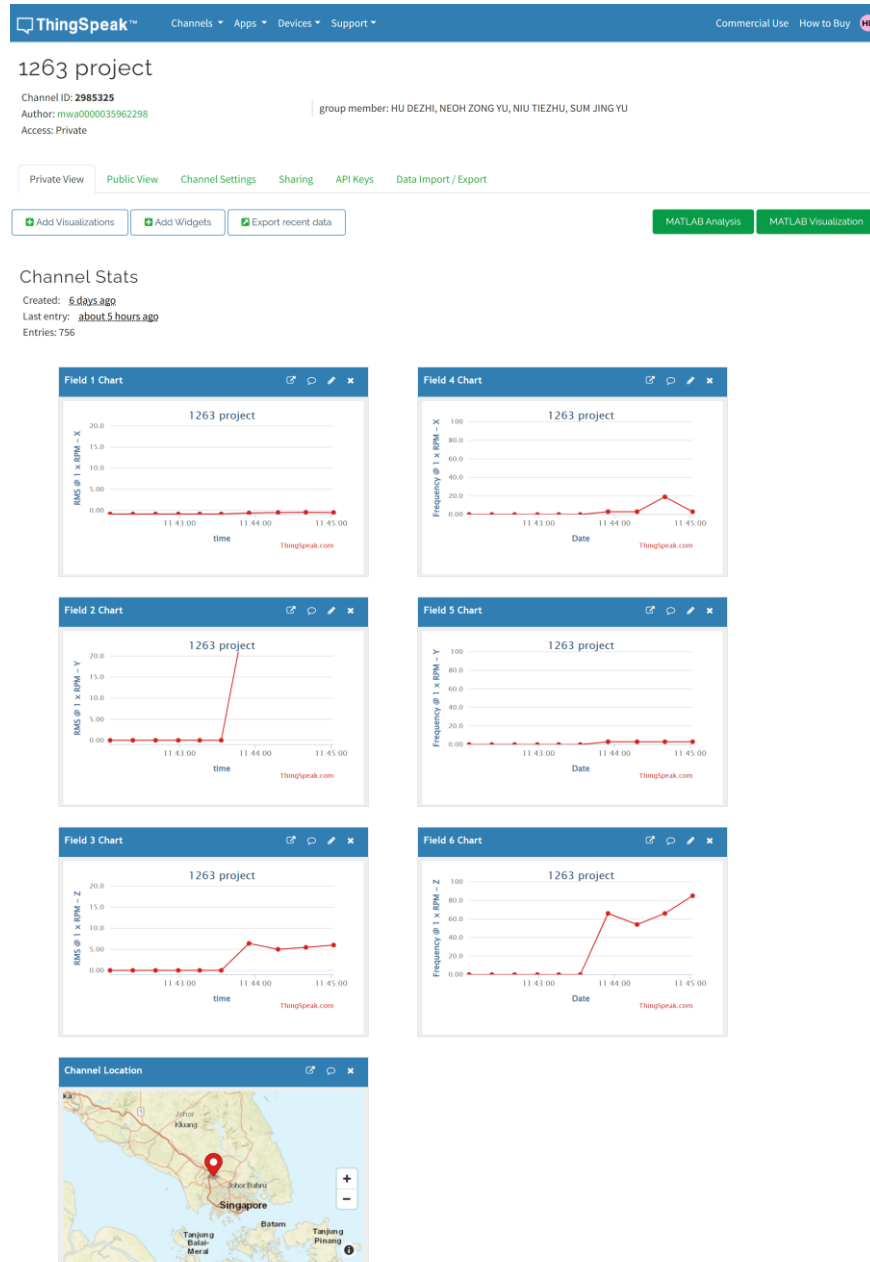


Figure 3.14: Dashboard Design for Data Visualization

3.4.2 Automated Notification and Alert Mechanism

Once the vibration level exceeds the predefined threshold of 3.0 g, an automatic email alert titled “1263 Alert: High Vibration Detected” is triggered and sent for notification, as illustrated in Figure 3.15. The email content includes the key vibration parameters at the time of threshold violation, specifically the values of RMS @ 1×RPM – X, Frequency @ 1×RPM – X, RMS @ 1×RPM – Y, Frequency @ 1×RPM – Y, and RMS @ 1×RPM – Z. By providing this real-time information, the system enables the operator to make prompt and informed decisions, such as inspecting the equipment or taking corrective action. This immediate response capability supports the implementation of predictive maintenance, allowing potential faults to be addressed before they escalate into major failures, thereby improving operational safety, reliability, and efficiency.

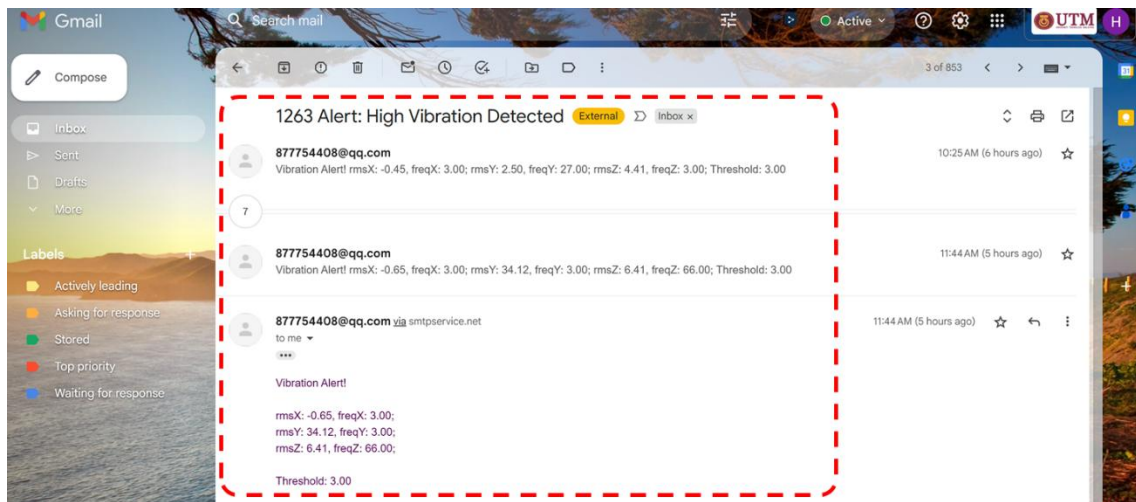


Figure 3.15: Example of alert notification sent through the email platform

3.5 Summary of Workflow

In general, the workflow of the wireless vibration condition monitoring system for the table fan is summarized in Figure 3.16. The process generally begins with the collection of mechanical vibration data using the transducer, last by the signal analysis in

both the time and frequency domains to identify any abnormal behavior based on the vibration patterns.

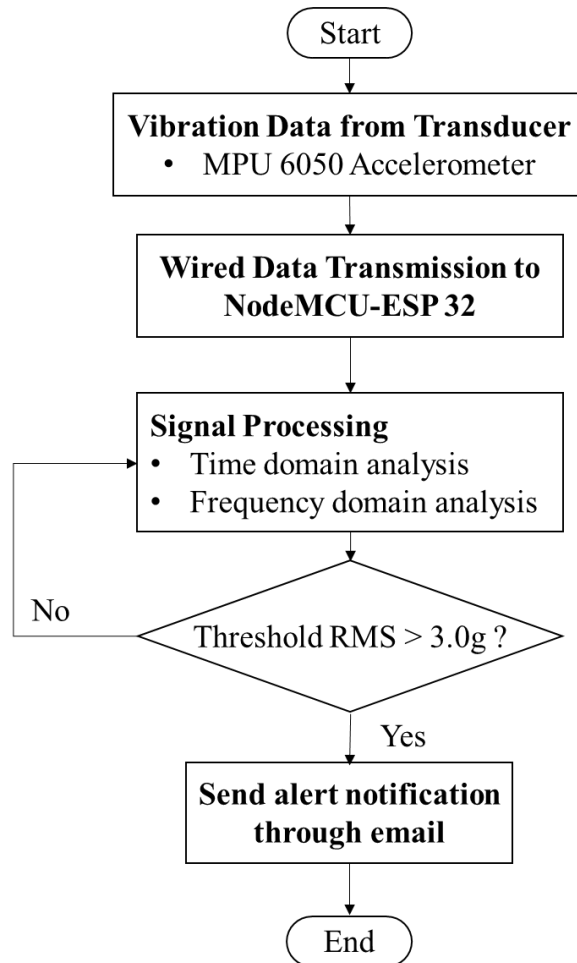


Figure 3.16: Summary workflow of the implemented system

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Introduction

This chapter presents and discusses the results obtained from the experimental setup used to develop the wireless vibration condition monitoring system for table fans. The system was then tested, and the data collected is categorized into three different scenarios: when the table fan is not running, when it is operating under normal conditions, and when it is operating under abnormal conditions. Therefore, a brief discussion is provided on the system output obtained under both conditions, highlighting the differences in vibration behavior and the system's ability to detect abnormalities.

4.2 Overview of Experimental Scenario

After the hardware and software components were fully implemented, the experimental boundary conditions were defined to simulate the vibration monitoring and fault detection process. The table fan was operated under three distinct conditions: normal operation and abnormal operation. To simulate the abnormal condition, which induces significant vibration, an external load was attached to one side of the fan blade. This setup creates an imbalance around the fan shaft axis, generating additional centrifugal force and resulting in increased vibration levels. Figure 4.1 illustrates the experimental boundary condition setup used for this investigation, while Table 4.1 displays the operating conditions for each table fan's status.

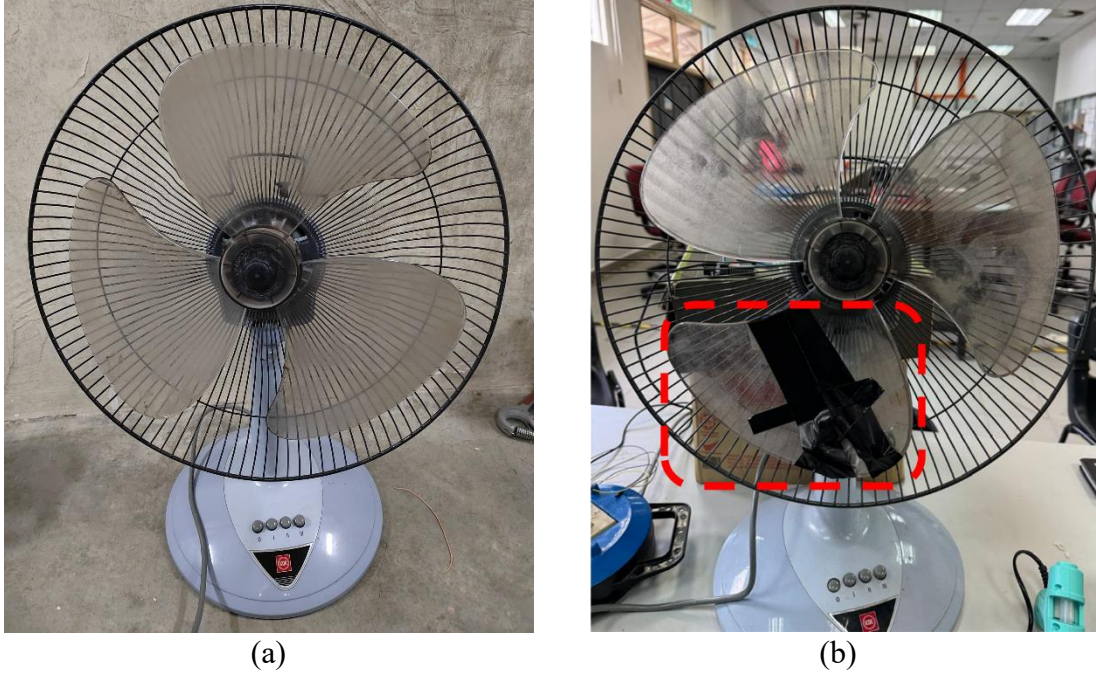


Figure 4.1: (a) Table fan setup for normal condition; (b) Table fan setup for abnormal condition

Table 4.1: Experimental scenario setup

Status	Fan Power (constant RPM)	External Load
No Run	OFF	NO
Normal Run	ON	NO
Abnormal Run	ON	YES

The “No Run” condition refers to the fan being in a non-operational state, which is used to record the baseline or background vibration, including any electronic noise from the sensor or the surrounding environment. In the “Normal Run” condition, the fan operates under normal, healthy conditions at a steady rotational speed (RPM), allowing the system to establish the typical vibration pattern in the absence of faults. During this phase, the RMS value must be closely monitored. If the observed RMS consistently exceeds the preset threshold, adjustment of the threshold is necessary to avoid false alarms. Finally, in the “Abnormal Run” condition, a simulated fault is introduced by attaching a

load to the fan blade to create an imbalance. This scenario is designed to evaluate the system's ability to detect changes in vibration when a fault is present.

4.3 Time-Domain Analysis

Figure 4.2 presents a line graph showing the RMS values of each independent axis plotted against time (in seconds), while Figure 4.3 displays the corresponding numerical RMS data.

Under the “No Run” condition, the RMS values for all three axes fluctuate between approximately 0.05 g and 0.5 g. This baseline vibration may result from electrical or magnetic interference caused by nearby equipment, which can induce high-frequency noise that the sensor interprets as acceleration. Additionally, airflow disturbances around the sensor could generate flow-induced vibrations, further contributing to the background noise observed during this idle state.

During the “Normal Run” condition, the RMS values for all three axis generally increased but remained below the threshold value of 3.0 g. This rise in RMS indicates a progressive increase in vibration energy. However, the levels are still within an acceptable operational range. Specifically, the RMS value for the X-axis stabilized at approximately 0.6 g, the Y-axis at 1.5 g, and the Z-axis at 0.3 g. These results suggest that, even under steady operating conditions, the Y-axis exhibits the highest vibration contribution. This could be attributed to the natural aging and wear of the table fan, potential sensor misalignment, or external factors such as electromagnetic interference.

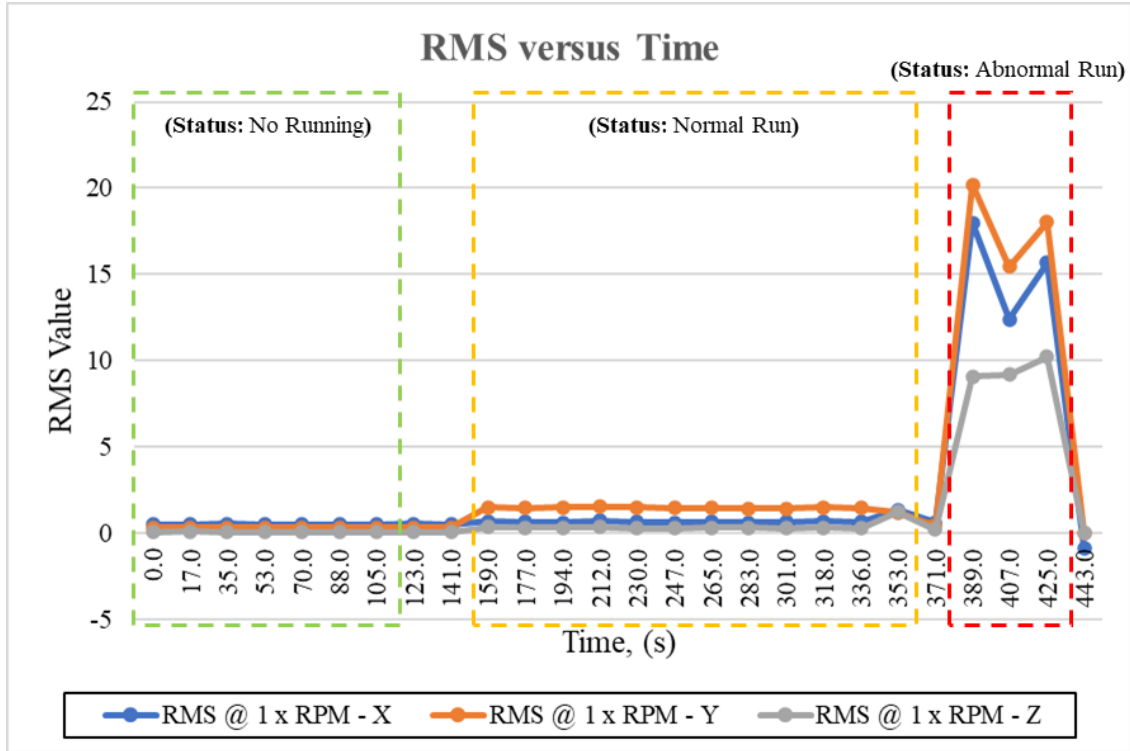


Figure 4.2: RMS value versus time with phase distribution

When the system shifts to the “Abnormal Run” condition, the RMS values increase sharply over time, indicating a sudden and progressive rise in vibration energy. This suggests that the table fan is experiencing a significant vibration issue. As shown in the table, the RMS values for all three axes rise considerably, eventually exceeding the threshold of 3.0 g. However, when comparing the magnitude of the RMS values across the three axes, the vertical axis (Z-axis) consistently exhibits the lowest RMS value. This indicates that the vibration is primarily concentrated in the horizontal (radial) directions of the table fan. This observation aligns with established vibration theory [12], where unbalance in rotating machinery predominantly induces radial vibrations, which are typically detected in the X and Y axes as the unbalanced mass generates centrifugal forces perpendicular to the shaft axis.

Status	Cumulative Time (s)	Time Diff. (s)	RMS @ 1 x RPM - X	RMS @ 1 x RPM - Y	RMS @ 1 x RPM - Z
No Run	0.00	-	0.503	0.295	0.053
	17.00	17.00	0.491	0.305	0.083
	35.00	18.00	0.505	0.295	0.053
	53.00	18.00	0.503	0.291	0.051
	70.00	17.00	0.492	0.290	0.050
	88.00	18.00	0.502	0.292	0.049
	105.00	17.00	0.504	0.291	0.045
	123.00	18.00	0.531	0.297	0.050
	141.00	18.00	0.504	0.292	0.062
Normal Run	159.00	18.00	0.668	1.488	0.329
	177.00	18.00	0.623	1.462	0.305
	194.00	17.00	0.607	1.488	0.290
	212.00	18.00	0.696	1.502	0.347
	230.00	18.00	0.619	1.477	0.275
	247.00	17.00	0.603	1.450	0.258
	265.00	18.00	0.621	1.458	0.285
	283.00	18.00	0.586	1.417	0.290
	301.00	18.00	0.616	1.415	0.275
	318.00	17.00	0.663	1.467	0.305
	336.00	18.00	0.630	1.427	0.282
	353.00	17.00	1.326	1.183	1.219
	371.00	18.00	0.582	0.406	0.205
Abnormal Run	389.00	18.00	17.967	20.177	9.069
	407.00	18.00	12.389	15.449	9.171
	425.00	18.00	15.680	18.029	10.209
	443.00	18.00	-0.880	0.000	0.000

Figure 4.3: RMS value for x, y, and z axis under different operating statuses

4.4 Frequency-Domain Analysis

Due to the absence of recorded vibration amplitude data, constructing the graph of vibration amplitude versus frequency is not feasible in this study. This limitation arose because the developed program did not extract and store the raw acceleration (amplitude) data during the initial experiment. Additionally, time constraints prevented a second round of data collection to address this omission. Therefore, alternative discussions and analyses are presented based on the available parameters, such as dominant or peak frequencies, to ensure meaningful insights can still be drawn from the experiment.

4.4.1 Frequency versus Time Plot

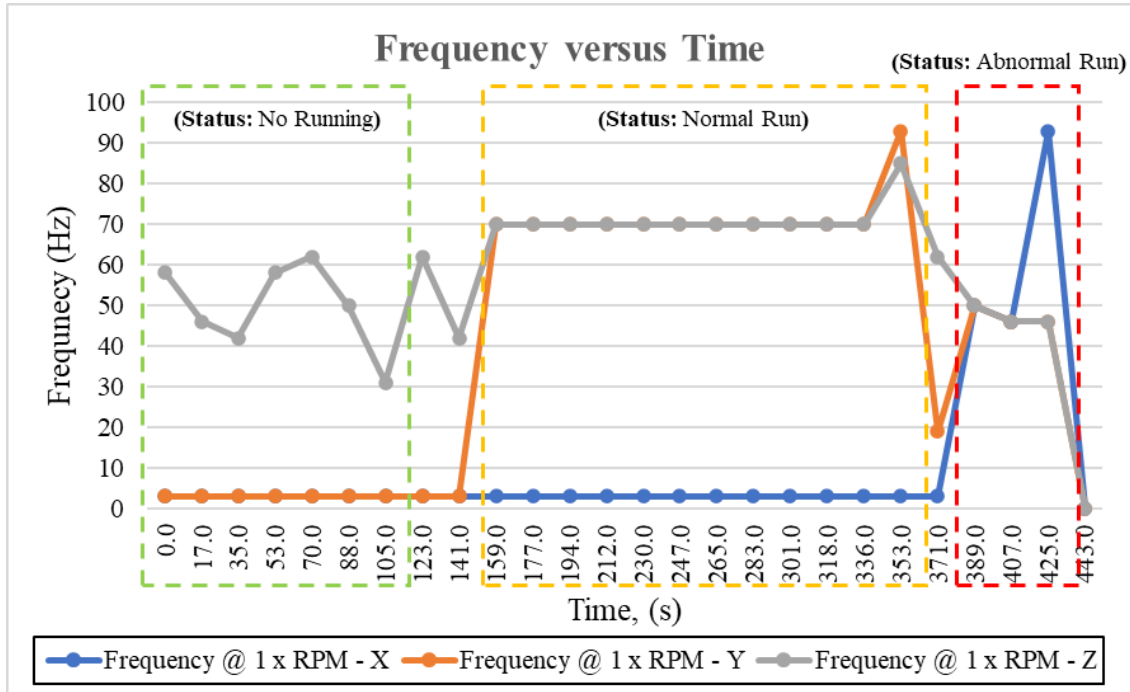


Figure 4.4: Frequency versus time plot

Status	Cumulative Time (s)	Time Diff. (s)	Frequency @ 1 x RPM - X	Frequency @ 1 x RPM - Y	Frequency @ 1 x RPM - Z
No Run	0.00	-	3.00	3.00	58.00
	17.00	17.00	3.00	3.00	46.00
	35.00	18.00	3.00	3.00	42.00
	53.00	18.00	3.00	3.00	58.00
	70.00	17.00	3.00	3.00	62.00
	88.00	18.00	3.00	3.00	50.00
	105.00	17.00	3.00	3.00	31.00
Normal Run	123.00	18.00	3.00	3.00	62.00
	141.00	18.00	3.00	3.00	42.00
	159.00	18.00	3.00	70.00	70.00
	177.00	18.00	3.00	70.00	70.00
	194.00	17.00	3.00	70.00	70.00
	212.00	18.00	3.00	70.00	70.00
	230.00	18.00	3.00	70.00	70.00
	247.00	17.00	3.00	70.00	70.00
	265.00	18.00	3.00	70.00	70.00
	283.00	18.00	3.00	70.00	70.00
	301.00	18.00	3.00	70.00	70.00
	318.00	17.00	3.00	70.00	70.00
	336.00	18.00	3.00	70.00	70.00
	353.00	17.00	3.00	93.00	85.00
Abnormal Run	371.00	18.00	3.00	19.00	62.00
	389.00	18.00	50.00	50.00	50.00
	407.00	18.00	46.00	46.00	46.00
	425.00	18.00	93.00	46.00	46.00
	443.00	18.00	0.00	0.00	0.00

Figure 4.5: Frequency for x, y, and z axis under different operating statuses

Figure 4.4 presents a line graph showing the frequency values of each independent axis plotted against time (in seconds), while Figure 4.5 displays the corresponding numerical frequency data.

Under the “No Run” condition, the vibration frequencies for both the X and Y axes remain stable at approximately 3 Hz, indicating minimal background vibration. However, the Z-axis frequency fluctuates between 31.0 Hz and 62.0 Hz. This baseline vibration may be attributed to electrical or electromagnetic interference from nearby equipment, which can induce high-frequency noise that the sensor interprets as false acceleration signals. Additionally, improper sensor tuning or mounting misalignment could also contribute to this noise in the Z-axis readings. However, since the RMS value for the Z-axis is low, it will still be considered acceptable.

During the “Normal Run” condition, the frequency values for all three axes generally stabilized. The X-axis frequency consistently remained at 3 Hz, while the Y-axis and Z-axis frequencies stabilized at approximately 70 Hz. This behaviour can be attributed to the orientation of the X-axis, which is aligned with the axial direction of the fan shaft. In rotating machinery, the axial direction typically experiences lower vibrational energy unless there are significant axial loads or shaft-related faults such as misalignment. In contrast, the Y and Z-axis capture vibrations in the radial directions, where most of the mechanical energy from rotating components is transmitted. The stability of the constant 70 Hz frequency indicates that the fan was operating at a steady rotational speed without significant fluctuations in load or mechanical behaviour. On the other hand, this frequency is likely associated with mechanical harmonics generated by the rotating blades or electromagnetic forces within the motor. The absence of noticeable frequency variations suggests a healthy operational state, with no emerging faults such as imbalance, looseness, or misalignment.

When the system transitions to the “Abnormal Run” condition, significant changes in frequency values are observed compared to the stable behavior during the Normal Run. The frequency along the X-axis, which was previously stable at 3 Hz under normal

operation, rises sharply to around 50 Hz and exhibits noticeable fluctuations. This sudden increase and instability suggest the presence of additional dynamic forces acting along the axial direction of the fan shaft, which are commonly associated with faults such as unbalance or mechanical looseness. Meanwhile, the frequencies along the Y and Z axes, which were stable at approximately 70 Hz during normal operation, shift to around 46 Hz and remain relatively steady. This reduction in frequency across both axes indicates a change in the system's dynamic behavior, possibly due to the added mass altering the rotational speed or influencing the natural frequencies of the fan. The stable but lower frequency reflects the fan's attempt to maintain rotational motion despite the imbalance, while the fluctuating axial frequency highlights the system's instability under the faulty condition.

4.5 Abnormality Detection and Alert System

When the operating condition shifts from “Normal Run” to “Abnormal Run,” the microcontroller continuously monitors the RMS values from all three axes (x, y, and z). If the RMS value of any axis exceeds the predefined threshold of 3.0 g, the system automatically triggers an alert notification in the form of an email to a designated recipient. This notification will continue to be sent as long as the table fan remains in abnormal condition. Figure 4.6 shows an example of alert information shown in the email platform, at which the alert information occurs at a cumulative time equal to 389.0 seconds.

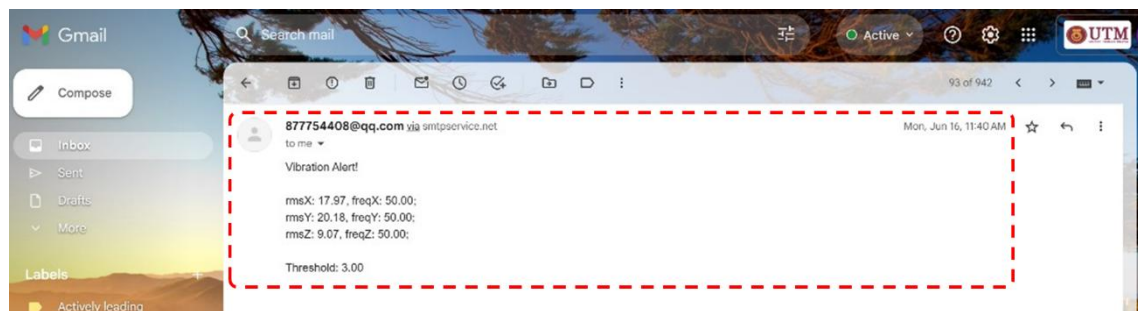


Figure 4.6: Alert information from the email platform

4.6 Summary of Findings and Limitations

In summary, the wireless vibration monitoring system effectively detected and distinguished vibration changes across different operating conditions: idle (No Run), normal operation, and abnormal operation. By performing both time-domain and frequency-domain analysis, the system leveraged key vibration indicators, specifically the RMS values and peak frequency detection, to reliably identify deviations from normal behavior that could signal the onset of mechanical faults. On top of that, the system utilized the RMS value as a primary threshold parameter for the fault detection methodology. When the calculated RMS value exceeded the predefined threshold, it would trigger the activation of the automated alert feature. The ESP32 microcontroller, equipped with built-in Wi-Fi capability, immediately transmitted the alert information to the designated cloud platform, ensuring that abnormal conditions were communicated in near real-time. This automated notification process enables timely intervention by maintenance personnel, thereby supporting the principles of predictive maintenance by detecting faults before they escalate into severe failures, reducing unplanned downtime, and improving overall equipment reliability.

However, due to the absence of raw acceleration (amplitude) data, the system was unable to record or display detailed vibration waveforms, which limited the ability to perform advanced vibration signature analysis, such as frequency spectra with amplitude scaling or detailed fault pattern recognition. This constraint prevented deeper insights into the nature of mechanical anomalies, such as identifying specific harmonics or transient shock events. Additionally, because the table fan has a lightweight and relatively flexible structure, even during normal operation, it exhibited comparatively high RMS values. This inherent sensitivity could increase the likelihood of false alarms unless the RMS threshold is carefully tuned and adjusted based on baseline operating conditions. Without proper calibration, minor vibrations that are typical for lightweight machinery could be misinterpreted as early signs of failure.

Furthermore, real-time data transmission in the system depends heavily on stable and frequent communication between the microcontroller and the cloud platform. Effective early fault detection requires the system to transmit data at short time intervals to ensure timely recognition of abnormal conditions and prompt issuance of alert notifications. However, in this project, the use of the free version of ThingSpeak imposed limitations on both the minimum update interval and data storage capacity. While the theoretical update limit is one data point every 15 seconds, the practical testing revealed an even slower actual transmission rate, with delays averaging around 18 seconds. This reduced time resolution may impact the responsiveness of the system, potentially causing delays in fault detection and alert delivery.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

The project successfully demonstrated the feasibility of developing a low-cost, wireless vibration monitoring system using the ESP32 microcontroller and MPU6050 accelerometer for a table fan. By implementing time-domain (RMS and kurtosis) and frequency-domain (FFT and peak frequency) analysis, the system effectively differentiated between idle, normal, and abnormal operational states of the fan. The integration of a cloud-based dashboard (ThingSpeak) and an automated email alert system provided real-time remote monitoring and fault notification capabilities, aligning with predictive maintenance goals.

The experimental results confirmed that the system could reliably detect abnormal vibration conditions, particularly those caused by mechanical imbalance, and trigger alerts when vibration levels exceeded predefined thresholds. However, limitations such as the absence of raw vibration waveform data and the delay in cloud data transmission (due to ThingSpeak's free version constraints) affected the depth and responsiveness of the system. Additionally, the system's sensitivity to lightweight equipment led to higher baseline RMS values, necessitating careful threshold tuning to avoid false positives.

To enhance the system's performance and applicability, in the future, it is recommended to incorporate raw data logging for advanced analysis, adopt faster and more robust cloud platforms for real-time responsiveness, and integrate machine learning algorithms for precise fault diagnosis. Expanding testing to industrial-grade machinery will validate scalability, while improving sensor mounting and calibration can reduce

noise and improve accuracy. Additionally, developing a mobile or web-based interface will increase user accessibility and engagement.

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